Problem 1

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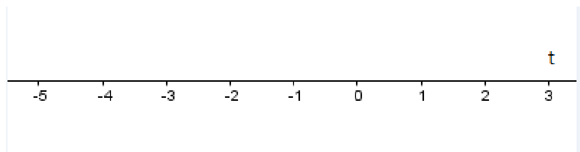
Problem 1 *The position, $s(t)$, of an object moving along a horizontal line is given by $s(t) = t^2 - 4$, where s is in meters and t is in seconds, $0 \leq t < 5$.*

(a) *Mark the position of the object on the line at time $t = 1$:*



(b) *Find the average velocity, v_{AV}, of the object during the time interval $[1, 3]$.*

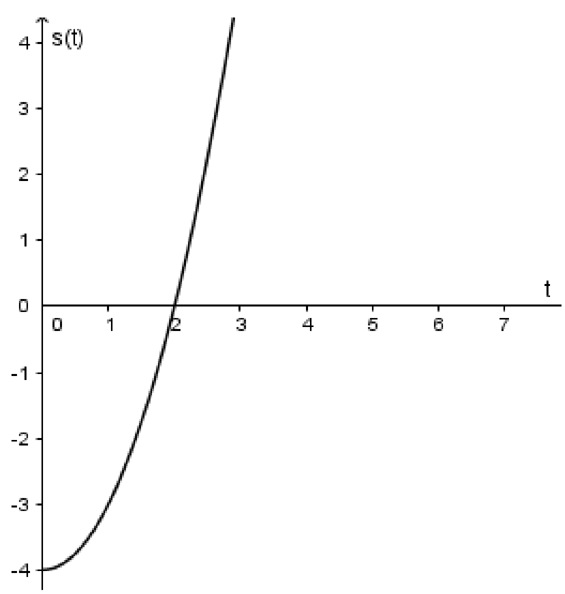
(c) *Compute the average velocity, $v_{\mathrm{AV}}(t)$, of the object during the time interval*

(i) $[1, t]$, *for* $1 < t < 5$;

(ii) $[t, 1]$, *for* $0 \leq t < 1$.

(d) *Find the instantaneous velocity, v_{inst}, of the object at $t = 1$. Justify your answer.*

(e) *The position-time graph of the function s is given in the figure below.*



(i) *Assume P is a point on the graph of s. Fill in the blank.*

$$P = (1, \underline{\hspace{1.5em}}).$$

(ii) *Plot the point P and draw the tangent line at this point in the figure above.*

(iii) *Find the slope, $m_{\tan}$, of the tangent line in part (ii). Explain.*